

LANE D — the descent: advance the derived half, scope the owed T2556 primitive (do NOT force it)

Tier discipline, cited to the LINE not the title (T2555 split, Cal §505 / K1541). The descent has two reductions with two different tiers; I advance the derived one and precisely *characterize* the owed one — I do not force it (forcing an owed primitive is exactly the over-claim this week kept catching).

The derived half — bulk-10 $\rightarrow$ boundary-5 (advance + state precisely)

DERIVED (T2555 reduction 1, §505): the totally-real maximal half-dimension theorem. D_{IV}^5 has real bulk dimension 10; its Šilov boundary $\check{S}$ is maximal totally-real, real dimension $5 = 10/2$. The reduction bulk-10 $\rightarrow$ boundary-5 drops the whole imaginary/Kähler part — the 5 “phase” directions of the holomorphic coordinates — and keeps the 5 “polarity” (real) directions. This is a genuine theorem (totally-real maximal submanifold + the Bergman reproducing property T349: the real boundary data determines the holomorphic interior). *Address*: (bulk interior, boundary $\check{S}$) read by verb (v) [boundary-orbit / dimension]. *Reading*: $\dim_{\mathbb{R}} \check{S} = n_C = 5$, half the bulk — target-innocent.

What it does and does not give (the honest boundary): it gives boundary-5, and it drops the *Kähler* phase (5 dimensions). It does not give the further boundary-5 $\rightarrow$ 4 step, and it does not drop the $S^1=SO(2)$ circle — that circle is *inside* the 5-dim boundary $(S^4 \times S^1)/\mathbb{Z}_2^{\wedge} \check{S}$, and dropping it is a *separate* reduction.

The owed half — boundary-5 $\rightarrow$ 4, the $S^1=SO(2)$ drop (scope, do NOT force)

STRUCTURAL, NOT DERIVED (T2555 reduction 2, §505 / K1541): the boundary-5 $\rightarrow$ 4 step drops the $S^1=SO(2)$ phase circle — one dimension, *between* the S^4 and S^1 factors. It is bridged only by the entry’s own naive one-bit-per-dimension count ($32 = 16 \times 2$), whose primitive is owed (T2556). $\Rightarrow$ the S^1 -drop must not serve as a derived constraint; it stays structural until T2556 lands.

Precisely what T2556 owes (declare the object; this is the deliverable, not a derivation): a derivation of the map $>$ “drop the $SO(2)$ phase circle” = “reduce boundary-5 $\rightarrow$ physical-4” that does not assume the one-bit-per-dimension count. Concretely, T2556 must supply *one* of: 1. a representation-theoretic reason the $SO(2)$ charge/phase is exactly the direction removed in passing to the $SO(3,1)$ Lorentz stratum of $SO(5,2) \supset SO(4,2) \supset SO(3,1)$ — i.e., the $SO(2)$ is the fibre of $\check{S} \rightarrow (4D \text{ base})$ under the conformal chain; or 2. a measure-theoretic reason the Szegő phase integral over S^1 contributes exactly one factor (the same S^1 whose round-trip gives the α^2 Born factor) — tying the drop to a counted integration, not a posited bit; or 3. a Jordan/Peirce reason the $c_1 \leftrightarrow c_2$ frame exchange ($\mathbb{Z}_2^{\wedge} \text{Peirce}$) pairs the two rank directions so that one $SO(2)$ survives as physical time — leaving 4D.

Each is a *candidate route*, none is claimed. The owed primitive is the identification of the $SO(2)$ -drop with the physical $5 \rightarrow 4$, target-innocently. Until one lands, the descent's 4D target is reached only through a structural step, and the ruler (one input, T2569) plus this structural S^1 -drop are the honest cost.

Tiers (stated)

step	tier
bulk-10 $\rightarrow$ boundary-5 (totally-real half-dim)	DERIVED (§505 reduction 1; T349)
boundary-5 $\rightarrow$ 4 ($S^1=SO(2)$ drop)	STRUCTURAL (§505 reduction 2; T2556 owed)
the ruler (scale)	DERIVED as one input (T2569 — necessity proved, shape not forced)

Lyra, 2026-08-21 (Lane D). Advanced the derived half: bulk-10 $\rightarrow$ boundary-5 = the totally-real maximal half-dimension theorem (drops the 5 Kähler-phase dims, keeps the 5 real; DERIVED, §505/T349), and stated precisely what it does NOT give (not the S^1 -drop, which is inside the 5-dim boundary). Scoped — did not force — the owed T2556 primitive: it must derive “drop $SO(2)$ = reduce $5 \rightarrow 4$ ” without the one-bit-per-dimension count; three candidate routes named (rep-theoretic fibre / Szegő-phase integral / Jordan frame), none claimed. S^1 -drop stays STRUCTURAL, cited to §505/K1541 not the T2555 title. Nothing pushed; CP existence-only.